Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.Tech, ME V semester
Name of the Paper: Operation research
Paper Code: TME-505
Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.

Q1.

a. What do you understand by Big M method? Explain by a suitable example. (CO1) (10 Marks)

OR

b. Write down the definition and history of Operation research. (CO1)

Q2.

a. What are the different applications of Operation research? Write down its limitations. (CO1) (10 Marks)

OR

a. Solve by simplex method: Maximize $Z=5x_1+4x_2$

Subject to:
$2x_1+3x_2\leq 12$
$x_1+x_2\leq 6$
$x_1\geq 0, x_2\geq 0$ (CO1)

Q3.

a. Maximize $Z=3x_1+2x_2$ (10 Marks)

Subject to:
$x_1+2x_2 \leq 4$
$3x_1+x_2 \leq 6$
$x_1\geq 0, x_2 \geq 0$ (CO1)

OR

b. Find out the total minimum cost of the following assignment problem:

		Machines			
		I	II	III	IV
Jobs	A	5	3	1	8
	B	7	9	2	6
	C	6	4	5	7
	D	5	7	7	6

(CO2)

Scanned with ACE Scanner

Mid Term (Odd) Semester Examination October 2024

Q4.

a. Find out initial solution by VAM: (10 Marks)

	To				Supply
From	7	3	8	6	60
	4	2	5	10	100
	2	6	5	1	40
Demand	20	50	50	80	

(CO2)

OR

b. Find out Optimal solution by MODI method. Initial basic solution is given in the table below:

	To				Supply
From	5	2(12)	4(1)	3(9)	22
	4	8	1(15)	6	15
	4(7)	6	7(1)	5	8
Demand	7	12	17	9	

(CO2)

Q5.

a. Find out initial solution by least cost method: (10 Marks)

	W1	W2	W3	Supply
F1	2	7	4	5
F2	3	3	1	8
F3	5	4	7	7
F4	1	6	2	14
Demand	7	9	18	

(CO2)

OR

b. Describe the different methods used in operation research.

(CO1)

Scanned with ACE Scanner